

FIGURE 12: SCIENTIFIC EVIDENCE & VALIDATION TAXONOMY MATRIX

Deterministic Classification of System Parameters: Measured, Validated, Calculated, Simulated & Demonstrated

[MEASURED]	[VALIDATED]	[CALCULATED]	[SIMULATED]	[DEMONSTRATED]
Direct Bench Instruments	Empirical Calibrations	Analytical Physics Models	Multiphysics Solvers	Physical System Builds
• Quiescent Current: 18.0 uA (Keithley 6514) • DSP FFT Latency: 2.49 ms (DWT Cycles) • MCU Die Temp: 25.4°C~26.8°C (TEMP) • Hardware Weight: 67 g (Digital Scale) • Enclosure Size: 65x55x15 mm (Calipers)	• Brood Core RTD: ±0.1°C NIST (TMP117) • Spatial Grid: ±0.5°C (5x DS18B20) • Audio Stream: 16 kHz 24-bit PCM • CUSUM Drift: -0.02°C/hr Queenless • Model 2 AI: 94.2% Acc (Zenodo) • Test Suite: 27/27 Tests Pass	• Packet Airtime: 18.2 ms (SX1262 Calc) • Daily Energy: 0.85 mWh/day (15m) • LoRa LOS Range: 4.2 km (Free-Space) • Link Margin: 26.16 dB @ 1 km • Apiary Duty Cycle: 0.202% (<1.0% Limit) • BOM Unit Cost: \$18.74 (No Custom PCB)	• Antenna Return Loss: S_11 = -22.4 dB (HFSS) • Antenna VSWR: 1.16 : 1 (50 Ohm Match) • Radiation Eff.: 91.4% (ABS Casing) • Thermal Core: 34.5°C (Fluent CFD) • Convective Plume: 0.18 m/s Natural Draft	• Transmitter Node: WisBlock RAK5005-O • Receiver Reader: RPI 3B+ & WAVESHARE • Wiring Terminals: 4:2 Levers & PG-7 • Local Server: FastAPI Daemon • Offline Database: SQLite WAL Store